

PAGE 2 — Income, marital status & children

Income

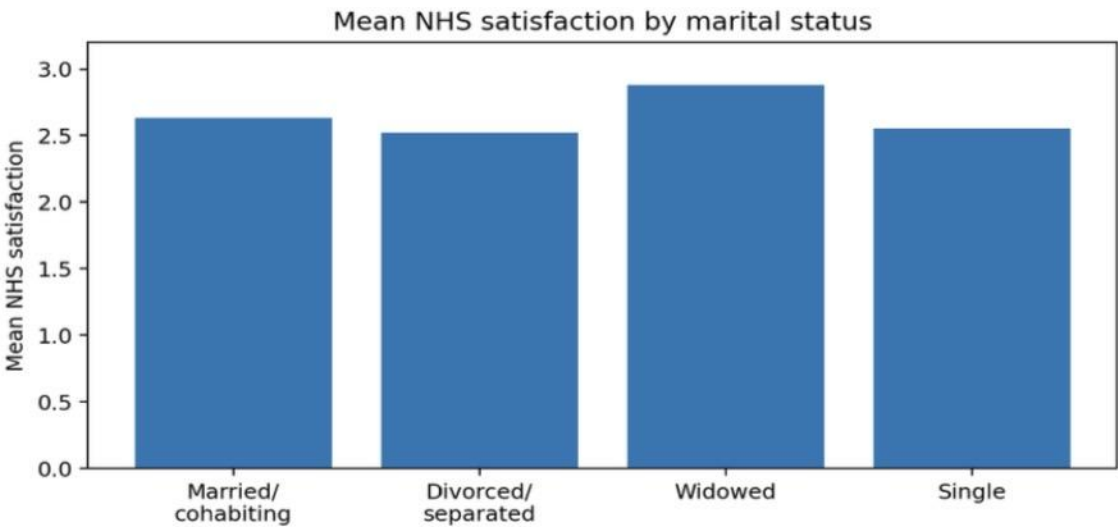
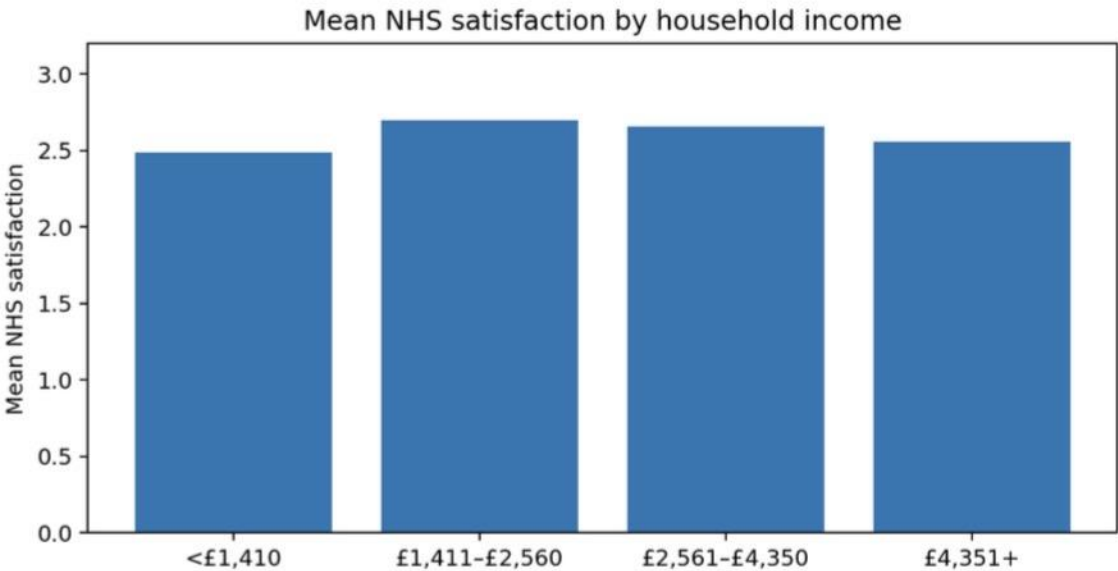
Mean satisfaction varies modestly across income groups. The highest mean is 2.70 for £1,411–£2,560; the lowest is 2.49 for <£1,410.

Marital status

Means range from 2.52 (divorced/separated) to 2.88 (widowed). The widowed estimate should be interpreted cautiously because N = 16.

Children

Respondents with children: 2.56; without children: 2.63. The difference is small.



PAGE 5 — Overall conclusions & limitations

Overall conclusion

Socio-demographic characteristics show only weak relationships with NHS satisfaction. Sex and education are not statistically significant, while age has virtually no explanatory power.

Policy implication

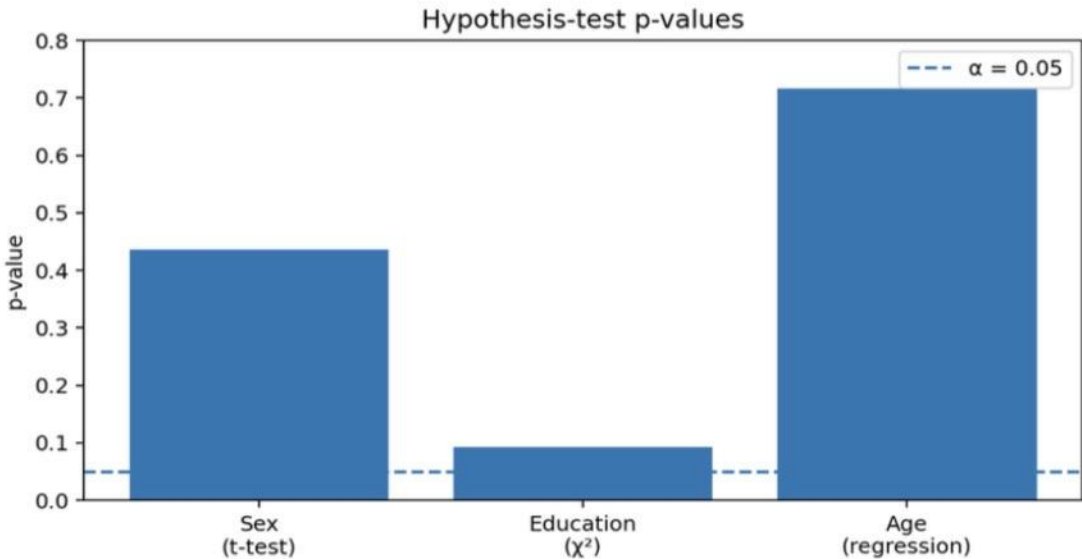
Dissatisfaction appears relatively diffuse rather than concentrated in one demographic group. Broad service-delivery improvements may therefore be more relevant than demographic targeting.

Limitations

The outcome is ordinal but treated as continuous; complete-case analysis may introduce bias; and cross-sectional data cannot establish causality.

Exam takeaway

Remember: BSA 2021; N = 3,307; $\alpha = 0.05$; sex $p = 0.437$; education $p = 0.093$; age $p = 0.717$; age $R^2 \approx 0$.



PAGE 3 — Sex & education

Sex

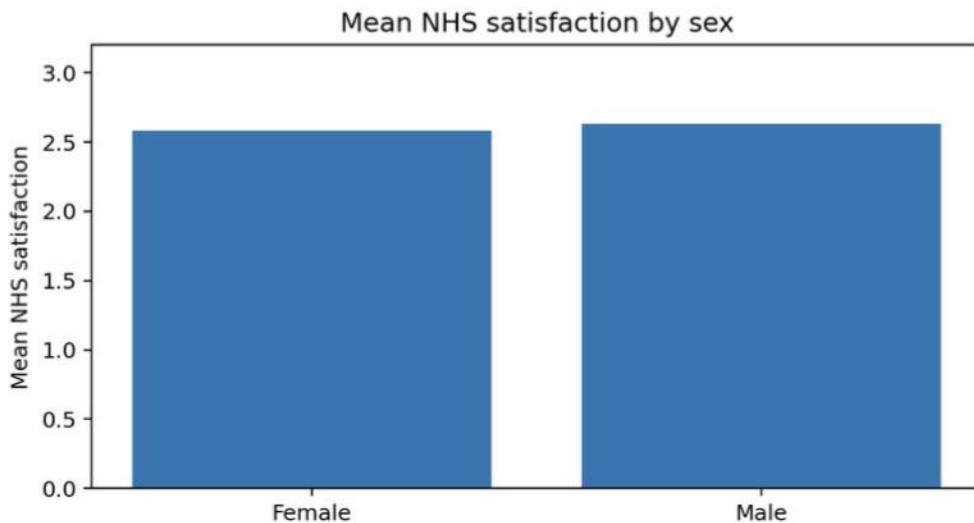
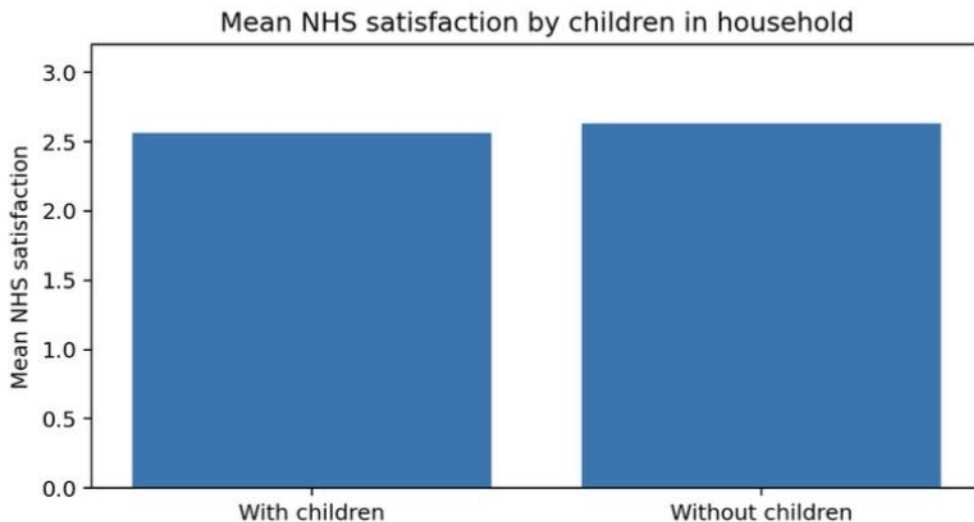
Female mean = 2.58; male mean = 2.63. $t = -0.78$, $df = 2888$, $p = 0.437$. There is no statistically significant difference.

Education

$\chi^2 = 10.84$, $df = 6$, $p = 0.093$. Since $p > 0.05$, there is no statistically significant association between education and NHS satisfaction.

Assumptions

Levene's test for the sex comparison: $p = 0.308$, supporting equal variances. Shapiro-Wilk $p < 0.001$ indicates non-normality, but the large sample makes the t-test reasonably robust.



PAGE 4 — Age & regression

Regression

Intercept = 2.56 ($p < 0.001$). Age coefficient = 0.00117 (SE = 0.00323; $p = 0.717$). $R^2 = 0.00004$; adjusted $R^2 \approx 0$.

Interpretation

Age has an extremely small positive coefficient and is statistically insignificant. The model explains virtually none of the variation in NHS satisfaction.

Diagnostics

Residual normality is violated (Shapiro-Wilk $p < 0.001$). No clear funnel pattern is observed, so there is no strong visual evidence of heteroscedasticity.

Important

The original materials do not provide the respondent-level age/satisfaction observations, so no invented scatterplot is included.

NHS Satisfaction — 5-Page Study Summary

PAGE 1 — Research question, data & methodology

Research question

The report examines whether NHS satisfaction differs across socio-demographic groups: household income, marital status, children in the household, sex, education and age.

Data

British Social Attitudes Survey 2021 (BSA 2021). Analytical sample: 3,307 respondents. NHS satisfaction is measured on a five-point ordinal scale and treated as continuous for most analyses.

Sample

Mean NHS satisfaction $\approx$ 2.60; approximately 50% female; around 42% live in households with children.

Methods

Descriptive statistics; independent-samples t-test for sex; Chi-square test of independence for education; simple linear regression for age.

Significance

All hypothesis tests use $\alpha = 0.05$. $p < 0.05$ = statistically significant; $p \geq 0.05$ = not statistically significant.